

Let $B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$ $AB = [Ab_1, Ab_2, Ab_3]$ (From the defn. of matrix multiplication)

$Ab_1 = \begin{bmatrix} 1 & -2 \\ -2 & 5 \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \end{bmatrix} = b_{11} \begin{bmatrix} 1 \\ -2 \end{bmatrix} + b_{21} \begin{bmatrix} -2 \\ 5 \end{bmatrix}$ (From the defn. of Ax)

$= \begin{bmatrix} b_{11} \\ -2b_{11} \end{bmatrix} + \begin{bmatrix} -2b_{21} \\ 5b_{21} \end{bmatrix}$ (By the scalar multiple of vector defn.)

$= \begin{bmatrix} b_{11} - 2b_{21} \\ 5b_{21} - 2b_{11} \end{bmatrix}$ (Vector sum defn.)

$Ab_2 = \begin{bmatrix} 1 & -2 \\ -2 & 5 \end{bmatrix} \begin{bmatrix} b_{12} \\ b_{22} \end{bmatrix} = \begin{bmatrix} b_{12} - 2b_{22} \\ -2b_{12} + 5b_{22} \end{bmatrix}$ (Similar calculation of Ab_1)

$Ab_1 = \begin{bmatrix} -1 \\ 6 \end{bmatrix}$ $\therefore b_{11} - 2b_{21} = -1$ and $-2b_{11} + 5b_{21} = 6$

$2b_{11} - 4b_{21} = -2$ $\therefore b_{21} = 4$

$b_{11} - 2 \times 4 = -1 \Rightarrow b_{11} = 7$

The 1st column of B : $\begin{bmatrix} 7 \\ 4 \end{bmatrix}$

$Ab_2 = \begin{bmatrix} 2 \\ -9 \end{bmatrix}$

$\therefore b_{12} - 2b_{22} = 2$ and $-2b_{12} + 5b_{22} = -9$

$2b_{12} - 4b_{22} = 4$

$b_{12} - 2 \times 8 = 2 \Rightarrow b_{12} = 18$

$\therefore b_{22} = -5$

The 2nd column of B : $\begin{bmatrix} 18 \\ -5 \end{bmatrix}$

18) Suppose the first two columns, b_1 and b_2 , of B are equal. What can you say about the columns of AB (if AB is defined)? Why?

Solution:

By the defn. of matrix multiplication, the 1st two columns of AB are Ab_1 and Ab_2 . We know the fact that each column of AB is a linear combination of the columns of A using weights from the corresponding column of B . $\therefore b_1$ and b_2 are equal, $\therefore Ab_1 = Ab_2$.

19) Suppose the third column of B is the sum of the first two columns. What can you say about the third column of AB ? Why?

Solution:

Let, b_1, b_2 and b_3 be the 1st 3 columns of B , where $b_3 = b_1 + b_2$ (Vector addition). By the defn. of Matrix multiplication, the first 3 columns of AB are Ab_1, Ab_2 and Ab_3 .